

AI-Assisted Heart Disease Prediction Project

Course: Artificial Intelligence | Final Examination

Scenario: HealthPlus Hospital needs an AI system to predict heart disease.

Dataset	UCI Cleveland Heart Disease dataset
Problem Type	Binary classification
Best Model	Random Forest
Best Accuracy	93.44%

1. Dataset Understanding

Target variable: target. It is made from target_raw. Value 0 means no heart disease and values 1-4 mean heart disease. This is a binary classification problem.

Numerical features: age, resting blood pressure, cholesterol, maximum heart rate, and oldpeak. Categorical features: sex, chest pain type, fasting blood sugar, resting ECG, exercise angina, slope, ca, and thal.

Important expected features are chest pain type, exercise angina, oldpeak, maximum heart rate, ca, thal, age, and slope. Two challenges are missing values in ca/thal and the small dataset size.

2. Exploratory Data Analysis

Item	Result
Raw rows and columns	303 rows, 14 columns
Clean rows and columns	303 rows, 15 columns
Missing values in raw data	6
Duplicate rows	0
Class distribution	{0: 164, 1: 139}

Class distribution means how many records belong to each target class. In this project, it shows how many patients have no heart disease and how many patients have heart disease. It is important because a highly unbalanced dataset can make a model look accurate while still missing the smaller class.

EDA plots show that the target is fairly balanced, older age and lower maximum heart rate are connected with disease, and chest pain type plus exercise angina are strong warning signs. The correlation matrix also shows useful relationships with oldpeak and thalach.



Figure 1: Class distribution. This chart compares patients with no heart disease and patients with heart disease. The classes are fairly balanced, so accuracy is more meaningful than it would be in a highly imbalanced dataset.

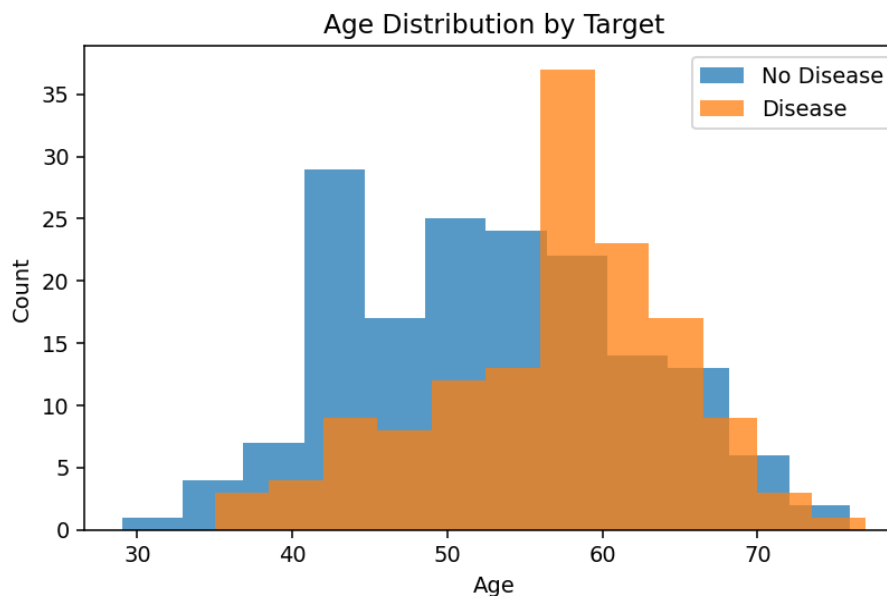


Figure 2: Age distribution by target. This chart shows how age is spread across both classes. Heart disease cases appear more common in older age groups, which means age is a useful risk-related feature.

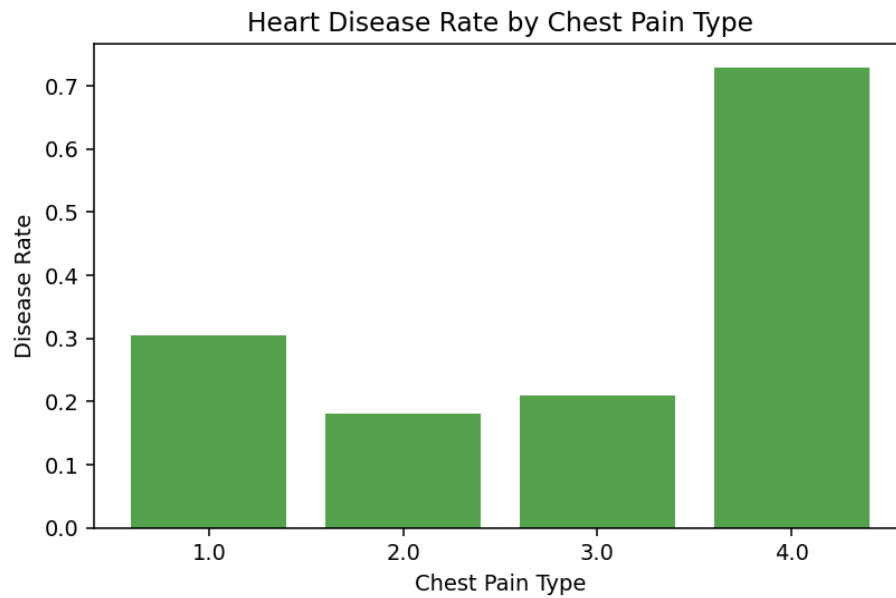


Figure 3: Heart disease rate by chest pain type. Some chest pain categories have a higher disease rate than others. This supports using chest pain type as an important categorical feature.

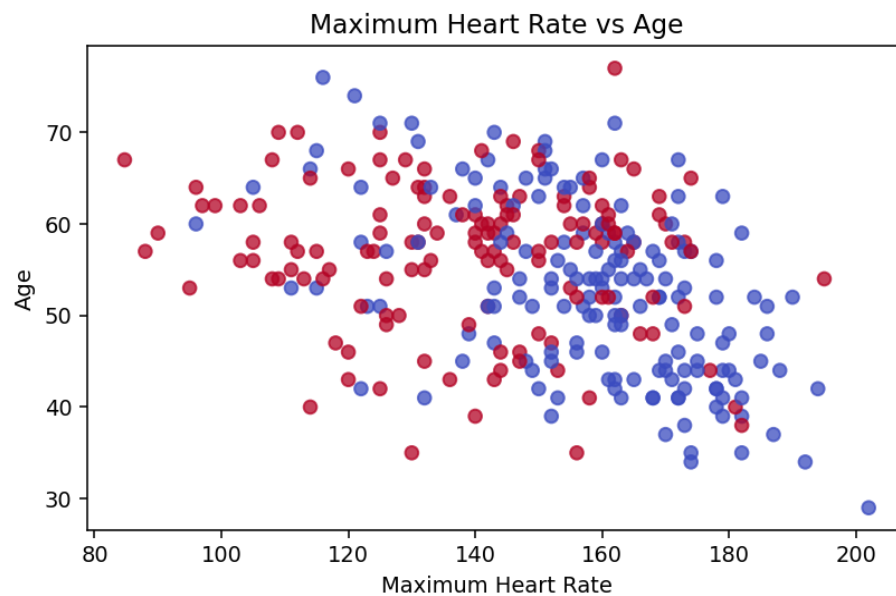


Figure 4: Maximum heart rate versus age. The scatter plot helps show how heart rate and age relate to the target. Lower maximum heart rate and higher age are often connected with higher risk.

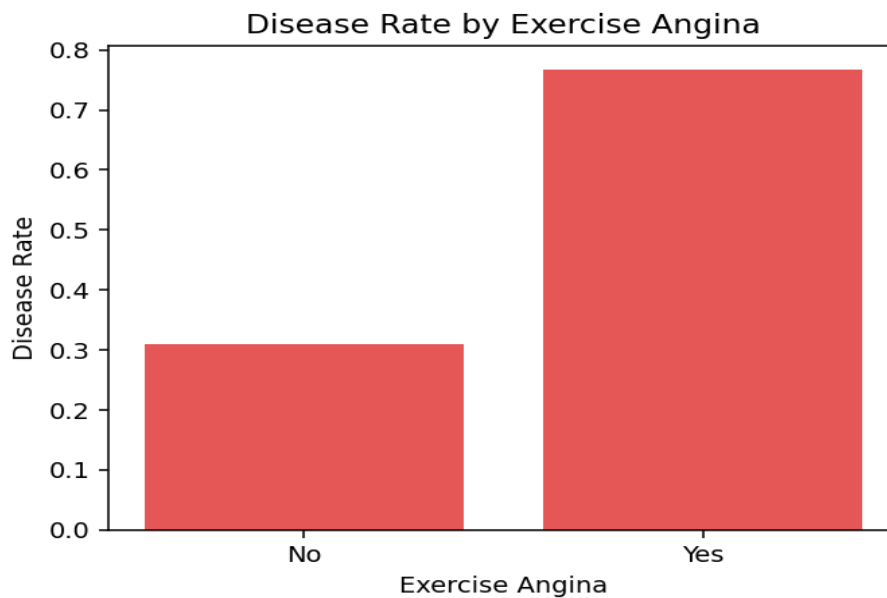


Figure 5: Disease rate by exercise induced angina. Patients with exercise induced angina show a higher heart disease rate. This makes exang an important warning feature.

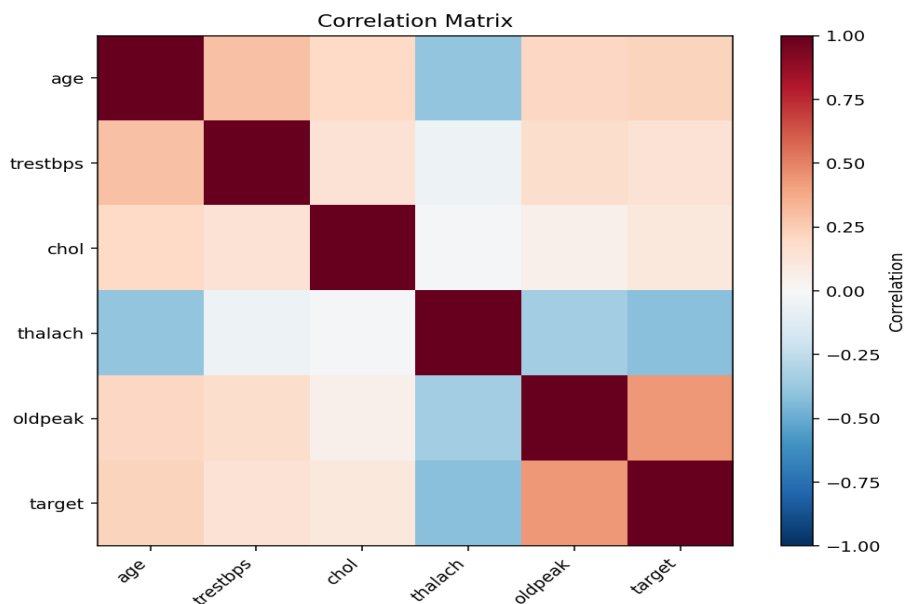


Figure 6: Correlation matrix. This heatmap shows relationships between numeric features and the target. It helps identify useful predictors and also checks whether features are strongly related to each other.

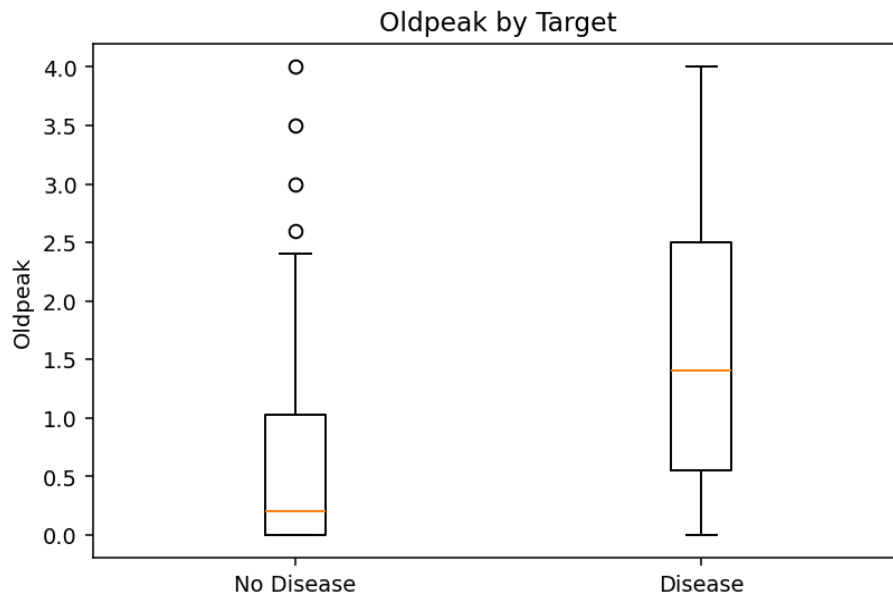


Figure 7: Oldpeak by target. This boxplot compares oldpeak values for disease and no disease patients. Higher oldpeak values are more common in heart disease cases, so oldpeak is an important medical signal.

3. Data Cleaning and Preprocessing

Missing numeric values were filled with the median because the median is less affected by extreme values. Missing categorical values were filled with the mode because it is the most common category.

Duplicates were removed. Numeric outliers were capped using the IQR method. IQR means Interquartile Range. Q1 means the first quartile, or the 25th percentile, so 25% of values are below Q1. Q3 means the third quartile, or the 75th percentile, so 75% of values are below Q3. IQR is calculated as $Q3 - Q1$. Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are treated as outliers and capped.

Categorical variables were one-hot encoded. Numeric variables were scaled because Logistic Regression and SVM work better when numeric values are on a similar scale.

4. Feature Engineering

Feature engineering means creating new useful columns from existing columns so the model can learn patterns more easily. It does not change the target answer; it only gives the model better input information.

New simple features were created: age_group, cholesterol_risk, bp_risk, heart_rate_gap, oldpeak_high, and risk_count. These features convert medical risk ideas into values that models can learn from.

Model	Baseline Accuracy	After Feature Engineering Accuracy
Logistic Regression	0.8689	0.8689
Random Forest	0.8689	0.9344
Support Vector Machine	0.8852	0.8689

5. Machine Learning Models

Logistic Regression: This model works like a simple scoring system. It gives weight to each feature, adds the scores, and then predicts the chance of heart disease. It is easy to understand and works well for many binary classification problems.

Random Forest: This model creates many small decision trees. Each tree makes a prediction, and the forest combines their votes. It often performs well because many trees reduce the risk of one weak tree making the final decision.

Support Vector Machine: SVM tries to find the best boundary that separates patients with heart disease from patients without heart disease. It is useful for medical classification because it can work well with smaller datasets.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8689	0.8333	0.8929	0.8621	0.9675
Random Forest	0.9344	0.9	0.9643	0.931	0.9724
Support Vector Machine	0.8689	0.8333	0.8929	0.8621	0.961

Accuracy means the percentage of total correct predictions. Precision means, when the model predicts heart disease, how often it is correct. Recall means how many real heart disease patients the model successfully finds. F1-score is a balance between precision and recall. ROC-AUC shows how well the model separates disease and no disease classes.

Best model selected: Random Forest. It was selected by comparing recall, F1-score, ROC-AUC, and accuracy. SVM was included because it is commonly useful for medical classification when the dataset is not very large and a clear decision boundary is needed. In a hospital problem, recall is very important because missing a patient with heart disease is risky.

6. Model Evaluation

A confusion matrix is a table that shows correct and wrong predictions. For binary classification it has true negatives, false positives, false negatives, and true positives. In this medical task, false negatives are dangerous because they mean a patient has heart disease but the model predicts no disease.

Logistic Regression confusion matrix: $\begin{bmatrix} 28 & 5 \\ 3 & 25 \end{bmatrix}$. Accuracy=0.8689, Precision=0.8333, Recall=0.8929, F1=0.8621, ROC-AUC=0.9675.

Random Forest confusion matrix: $\begin{bmatrix} 30 & 3 \\ 1 & 27 \end{bmatrix}$. Accuracy=0.9344, Precision=0.9, Recall=0.9643, F1=0.931, ROC-AUC=0.9724.

Support Vector Machine confusion matrix: $\begin{bmatrix} 28 & 5 \\ 3 & 25 \end{bmatrix}$. Accuracy=0.8689, Precision=0.8333, Recall=0.8929, F1=0.8621, ROC-AUC=0.961.

Most appropriate metric: Recall, with F1-score also important. Recall reduces false negatives, while F1-score balances recall and precision.

7. Model Interpretation

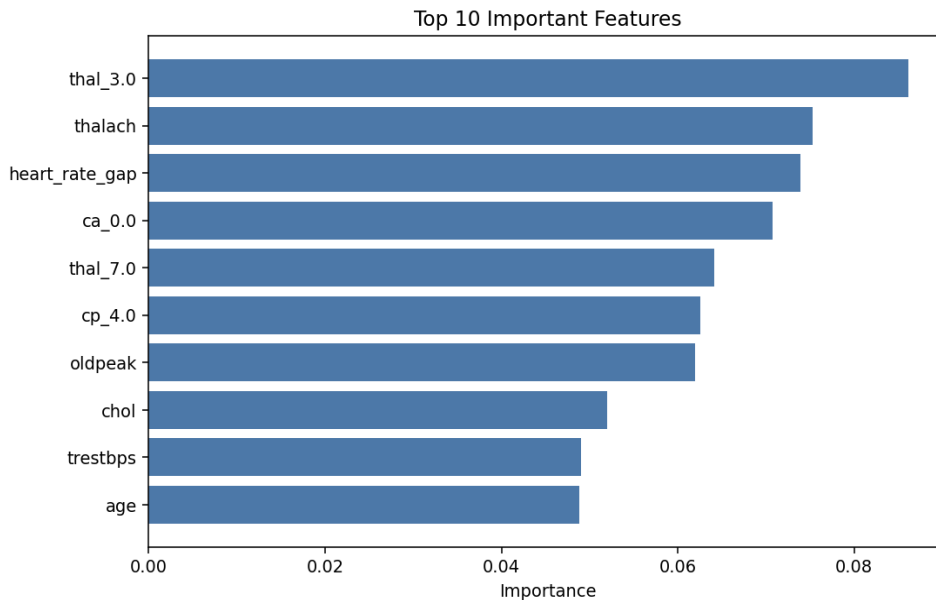


Figure 8: Feature importance. This chart ranks the top features used by the best model. Features with higher importance influenced the model prediction more strongly.

Feature	Importance
thal_3.0	0.0862
thalach	0.0753
heart_rate_gap	0.0739
ca_0.0	0.0707
thal_7.0	0.0641
cp_4.0	0.0626
oldpeak	0.062
chol	0.052
trestbps	0.049
age	0.0488

The top features are useful because they describe chest pain, exercise response, blood vessel information, heart rate, and ST depression. These are medically related to heart disease risk.

8. AI-Assisted Development Reflection

Prompt 1

Prompt used: Read the exam instruction PDF and list all required tasks and submission files.

AI output and my modification: AI identified the nine tasks and required files. I followed those sections in the script, notebook, report, and prompt history.

Prompt 2

Prompt used: Find a proper heart disease dataset for this project and explain the target variable.

AI output and my modification: AI suggested the UCI Cleveland Heart Disease dataset. I used the official UCI data and converted target_raw into binary target.

Prompt 3

Prompt used: Write easy Python code for cleaning, EDA, feature engineering, Logistic Regression, Random Forest, SVM, and evaluation.

AI output and my modification: AI suggested pandas, matplotlib, and scikit-learn. I used SVM because this is a medical classification task and kept pipelines for fair comparison.

Prompt 4

Prompt used: Create a short final report with section-by-section explanation, model results, prompts, limitations, ethics, and recommendations.

AI output and my modification: AI helped organize the report. I filled the report with actual results from the model run.

AI mistake noticed: AI first suggested some extra complex ideas, but I kept the code simple for an exam project.

My improvements: I used an official data source, added SVM for medical classification, checked the metrics from real model output, and added a separate AI prompt history file.

9. Critical Discussion and Recommendations

Limitations: the dataset is small and old, so results may not generalize to all hospital patients. Bias can happen if the patient group does not represent the real population. Data leakage was avoided by splitting train/test before model fitting inside pipelines.

Ethics and privacy: patient data must be protected, anonymized, and used only with permission. The model should support doctors, not replace doctors. For deployment, the hospital should validate the model on newer local patient data before real use.